Anamaria Crisan, PhD

Lead Research Scientist
Tableau Research
@amcrisan | https://amcrisan.github.io | https://www.linkedin.com/in/amcrisan/

I am a Lead Research Scientist at Tableau, where I am part of the Office of Chief Technology Officer (CTO). I have 10 years of experience translating applied machine learning (ML) research into products and policy making within industry, government, and academia. My research track record includes award-winning publications at to top-tier conferences of the ACM (CHI), IEEE (TVCG, CG&A) and biomedical journals (Nature, Oxford Bioinformatics, PLOS).

Research Interests*:* Interactive Machine Learning, Human Centered ML/AI, Data Science, Data Visualization, Bioinformatics

Education

2015 - 2019 **PhD Computer Science**, *University of British Columbia*
Thesis: Visualizing Heterogenous Data in Genomic Epidemiology
Supervisors: Dr. Tamara Munzner and Dr. Jennifer L. Gardy
Committee: Dr. Raymond Ng and Dr. Bonnie Henry
🏆 *Notable Awards*: Canada Vanier Scholar and UBC Public Scholar

2008 - 2010 **MSc Bioinformatics**, *University of British Columbia*
Thesis: Mutation Discovery in Regions of Segmental Cancer Genome Amplifications from Next Generation Sequencing of Tumours.
Supervisors: Dr. Sohrab Shah and Dr. Samuel Aparicio
🏆 *Notable Awards*: CHIR-MSFHR Bioinformatics Trainee Award

2004 - 2008 **BSc Computer Science with Honours**, *Queen's University at Kingston*
Specialization: Biomedical Computing
Honours Thesis: Genomics of Autism
Honours Thesis Supervisor: Dr. Hagit Shatkay
🏆 *Notable* Awards: NSERC USRA, CRA-W Distributed Mentorship

Research Employment

2019 - Present **Lead Research Scientist**, Tableau Research
Seattle Washington, USA

- Lead of Human-ML/AI (HAI) collaboration strategic initiative and co-lead of Responsible ML/AI Strategy Area
- Facilitated tactical strategy discussion to get buy-in for HAI investment
- Advocated for and managed development of a research data resource for analyzing workbooks that is now used across multiple orgs.
- Conducted analysis using statistical and ML methods for structured and unstructured (text) datasets derived from visual analytic workbooks
- Contributor to Tableau Main (Overcoming Misinformation, Data Science Reflection) and Engineering Blog (Interactive Machine Learning, AutoML and People)

2013 - 2015 **Research Scientist**, British Columbia Centre for Disease Control
Vancouver, British Columbia, CANADA

- Routinely collaborated with doctors, nurses, health officers, and statisticians to study, model, and monitor spread disease using genomic technology + ML techniques
- Developed data visualization tools to help domain experts get started with genomic + electronic health record data more effectively. Developed the reportable disease dashboard prototype that now appears on the BCCDC website.
- Developed and deployed ML models for improved water quality measures from metagenomic signatures from wastewater sampling
- Developed a clinical report for TB that has been adopted by global health care agencies
- Routinely invited to give talks about my work, and its findings, at medical grand rounds, to Canadian government officials, and also global public health agencies
- Conducted statistical analysis of data, including linear (and hierarchical) modelling, time series analysis, spatio temporal analysis, causal analysis, clustering, supervised and unsupervised learning

2010 - 2013 **Research Scientist**, Decipher Biosciences
Vancouver, British Columbia, CANADA

- Start-up (then GenomeDx Biosciences) where I was employee 3, grew it up to ~100 people, and which sold recently with a $600 million valuation.
- I was the co-inventor of the flagship product : a genomic classifier that predicts metastatic prostate cancer post-surgery. Lead the translation from research to production
- I participated in a series A funding round valued at around $12 million; I worked directly with investors on the technical due diligence
- I directly to contributed research and clinical study development for our portfolio for medicare reimbursement
- Collaborated with clinical and research partners to conduct statistical analysis of data, design, and analysis of clinical studies

Publications

My work appears across disciplines in biomedicine and computer science. Medical disciplines prioritize journal publication, whereas computer science disciplines prioritize conference proceedings. I have a total of 24 peer-reviewed publications with 13 as first author.

My work has appeared in Nature and Bioinformatics, as well top-tier venues in Computer Science, including CHI and TVCG. It has been cited nearly 3,700 times. Updated citation counts are available on google scholar. Included below is the current impact factor and acceptance rate for journals and conferences, respectively.

In what appears below:

Crisan, A = first or senior author publication

Crisan, A★ = joint first author

First, last † = mentee

(pdf), (blog), (code), (video,) and (draft) are all clickable links.

Submitted and In Preparation

2022 "Speaking my truth": affordances and expectations of semantic interactions for human guided machine learning.
Crisan A Shang M, Brochu E.
In preparation for TVCG
(draft)

2022 AutoML Trace : Surfacing and Tracing artifacts in automated data science workflows.
Rogers, J†, **Crisan A**
Submitted CSCW'22
(draft)

Peer Review Journal and Conference Publications

2021 Finding their data voice: Practices and challenges of dashboard users.
Tory, M, Bartram, L, Fiore-Gartland, B, **Crisan A**.
IEEE Computer Graphics and Applications [IF: 2.8]

2021 GEViTRec: Data reconnaissance through recommendation using a domain-specific visualization prevalence design space.
Crisan A, *Fisher, S†*, Gardy, JL, Munzner T.
IEEE Transaction on Visualization and Computer Graphics [IF: 4.5]
(pdf)

2021 Fits and Starts: Enterprise use of AutoML and the role of humans in the loop.
Crisan A, Fiore-Gartland, B
Proc. CHI'21 [acceptance rate: 25%]
(pdf)(blog)(video)
Best Paper Honorable Mention at CHI'21

2021 User ex Machina: Simulation as a design probe in human in the loop text analytics.
Crisan A, Correll M
Proc. CHI'21 [acceptance rate: 25%]
(pdf)(blog)(video)

2020 Passing the data baton: A retrospective analysis on Data Science work and workers.
Crisan A, Fiore-Gartland, B, Tory, M.
IEEE Transactions on Visualization and Computer Graphics [IF: 4.5]
(pdf)(blog)(video)
Best Paper at the Visualization in Data Science Symposium

2020 Divining insights: Visual analytics through cartomancy
McNutt, A. **Crisan A**, Correll, M.
ACM CHI'20 – alt.CHI
(pdf)

2019 Uncovering data landscapes through data reconnaissance and task wrangling
Crisan A, Munzner T.
IEEE Visualization Conference (Short Paper) [acceptance rate: 30%]
(pdf)

2018 A systematic method for surveying data visualizations and its resulting Genomic Epidemiology Visualization Typology: GEViT
Crisan A, Gardy JL, Munzner T
Oxford Bioinformatics [IF: 5.48]
(pdf)

2018 Adjutant: an R-based tool to support topic discovery for systematic and literature reviews
Crisan A, Munzner T, Gardy JL
Oxford Bioinformatics (Application Note) [IF: 5.48]
(pdf)(code)(blog)

2018 Evidence-based design and evaluation of a whole genome sequencing clinical report for the reference microbiology laboratory
Crisan A, McKee G, Munzner T, Gardy JL
PeerJ (Peer Reviewed) [IF: 2.12]
(pdf)

2016 Genomic analysis of a serotype 5 streptococcus pneumoniae outbreak in British Columbia, Canada, 2005 - 2009
Miller RR, Langille MG, Montoya V, **Crisan A**, Stefanovic A, Martin I, Hoang L, Patrick DM, Romney M, Tyrrell G, Jones SJ, Brinkman FS, Tang P
Canadian Journal of Infectious Diseases and Medical Microbiology [IF: 0.99]
(pdf)

2016 Declaring a tuberculosis outbreak over with genomic epidemiology
Hatherell H, Didelot X, Pollock SL, Tang P, **Crisan A,** Johnston JC, Colijn C, Gardy JL
Microbial Genomics [IF: NA]
(pdf)

2015 Spatio-temporal analysis of tuberculous infection risk among clients of a homeless shelter during an outbreak
Crisan A, Wong HY, Johnston JC, Tang P, Colijn C, Otterstatter M, Hiscoe L, Parker R, Pollock SL, Gardy JL
The International Journal of Tuberculosis and Lung Disease [IF: NA]
(pdf)

2015 Combined value of validated clinical and genomic risk stratification tools for predicting prostate cancer mortality in a high-risk prostatectomy cohort
Cooperberg MR, Davicioni E, **Crisan A,** Jenkins RB, Ghadessi M, Karnes RJ
European Urology [IF: 13.94]
(pdf)

2014 Clinical and genomic analysis of metastatic prostate cancer progression with a background of postoperative biochemical recurrence
Crisan A★, Alshalalfa M★, Vergara IA, Ghadessi M, Buerki C, Erho N, Yousefi K, Sierocinski T, Haddad Z, Black PC, Karnes RJ, Jenkins RB, Davicioni E
British Journal of Urology International [IF: 4.39]
(pdf)

2014 A genomic classifier predicting metastatic disease progression in men with biochemical recurrence after prostatectomy
Ross AE, Feng FY, Ghadessi M, Erho N, **Crisan A**, Buerki C, Sundi D, Mitra AP, Vergara IA, Thompson DJ, Triche TJ, Davicioni E, Bergstralh EJ, Jenkins RB, Karnes RJ, Schaeffer EM
Prostate Cancer and Prostatic Diseases [IF: 3.80]
(pdf)

2013 Validation of a genomic classifier that predicts metastasis following radical prostatectomy in an at risk patient population
Karnes RJ, Bergstralh EJ, Davicioni E, Ghadessi M, Buerki C, Mitra AP, **Crisan A**, Erho N, Vergara IA, Lam LL, Carlson R, Thompson DJ, Haddad Z, Zimmermann B, Sierocinski T, Triche TJ, Kollmeyer T, Ballman KV, Black PC, Klee GG, Jenkins RB
The Journal of Urology [IF: 5.16]
(pdf)

2013 Validation of a Genomic Classifier That Predicts Metastatic Disease Progression in Men With Biochemical Recurrence Post-Radical Prostatectomy
Feng FY, Ghadessi M, Davicioni E, **Crisan A**, Erho N, Mitra AP, Triche TJ, Jenkins RB, Ross AE, Schaeffer EM
International Journal of Radiation Oncology [IF: 5.16]
(pdf)

2013 Discovery and validation of a prostate cancer genomic classifier that predicts early metastasis following radical prostatectomy
Crisan A★, Erho N★, Vergara IA, Mitra AP, Ghadessi M, Buerki C, Bergstralh EJ, Kollmeyer T, Fink S, Haddad Z, Zimmermann B, Sierocinski T, Ballman KV, Triche TJ, Black PC, Karnes RJ, Klee G, Davicioni E, Jenkins RB
PLOS ONE [IF: 2.77]
(pdf)

2012 The clonal and mutational evolution spectrum of primary triple-negative breast cancers
Shah SP, Roth A, Goya R, Oloumi A, Ha G, Zhao Y, Turashvili G, Ding J, .., **Crisan A**, ... Marra MA, Aparicio S
Nature [IF: 40.14]
(pdf)
I was responsible for architecting and engineering the distributed data analysis infrastructure.

2012 JointSNVMix: a probabilistic model for accurate detection of somatic mutations in normal/tumour paired next-generation sequencing data
Roth A, Ding J, Morin R, **Crisan A**, Ha G, Giuliany R, Bashashati A, Hirst M, Turashvili G, Oloumi A, Marra MA, Aparicio S, Shah SP
Oxford Bioinformatics [IF: 5.48]
(pdf)

2012 Genomic dark matter in prostate cancer: exploring the clinical utility of ncRNA as biomarkers
Vergara I, Erho N, Triche T, Ghadessi M, **Crisan A**, Sierocinski T, Black PC, Buerki C, Davicioni E.
Frontiers in Genetics [IF: 3.79]
(pdf)

2012 Mutation discovery in regions of segmental cancer genome amplifications with CoNAn-SNV: a mixture model for next generation sequencing of tumors
Crisan A, Goya R, Ha G, Ding J, Prentice LM, Oloumi A, Senz J, Zeng T, Tse K, Delaney A, Marra MA, Huntsman DG, Hirst M, Aparicio S, Shah SP
PLOS ONE [IF: 2.77]
(pdf)

2010 SNVMix: predicting single nucleotide variants from next-generation sequencing of tumors
Goya R, Sun MG, Morin RD, Leung G, Ha G, Wiegand KC, Senz J, **Crisan A**, Marra MA, Hirst M, Huntsman D, Murphy KP, Aparicio S, Shah SP
Oxford Bioinformatics [IF: 5.48]
(pdf)

Workshop Publications

2021 Natto: Rapid and Visual Iteration of Analytic Data Models with Intelligent Assistance
Crisan,A, Setlur V.
VDS @ ACM KDD'21 and IEEE VIS'21
(pdf)

2018 How to evaluate an evaluation study? Comparing and contrasting practices in vis with those of other disciplines
Crisan A, Elliott M.
BELIV 18 (an IEEE VIS 2018 affiliated workshop)
(pdf)

2016 On regulatory and organizational constraints in visualization design and evaluation
Crisan A, Gardy JL, Munzner T
BELIV 16 (an IEEE VIS 2016 affiliated workshop)
(pdf)

Intellectual Property

2021 ***Cancer diagnostics using biomarkers (#US20140066323A1)***
Buerki C, Crisan A, Davicioni E, Erho NG, Ghadessi M, Jenkins RB, Vergara IA
Status: Granted as of 2021-06-15

2018 ***Prostate cancer prognostics using biomarkers (#EP2885640B1)***
Buerki C, Crisan A, Davicioni E, Erho NG, Ghadessi M, Jenkins RB, Vergara IA
Status: Granted as of 2018-07-18

Selected Posters

2018 Visualizing public health data in the era of microbial genomic epidemiology
Crisan A, Gardy JL, Munzner T
Amazon Research Days (Theme: AI & Society) – Seattle, USA

2018 Creating explorable visualization design spaces for domain experts: an example from infectious disease genomic epidemiology
Crisan A, Gardy JL, Munzner T
IEEE VIS 2018 - Berlin, Germany

2018 Visualizing public health data in the era of microbial genomic epidemiology
Crisan A, Gardy JL, Munzner T
National Canadian Student Health Research Forum – Winnipeg, Canada
Winner: CIHR GOLD Category - National Poster Competition

2017 Enabling systematic analysis of pathogen genomics data visualizations through GEViT: a genomic epidemiology visualization typology
Crisan A, Gardy JL, Munzner T
ASM NGS – Washington DC, USA

2017 Showcasing the design study methodology through simpler design challenges: An application to a microbial genomics clinical report design
Crisan A, Gardy JL, Munzner T
IEEE VIS 2017 – Phoenix, USA

Selected Talks

Selected Refereed Talks

August 2021 Natto: Rapid Visual Iteration of Analytic Data Models with Intelligent Assistance
ACM KDD'21 – VDS – Singapore (virtual due to COVID-19)

October 2021 Fits and Starts: Enterprise Use of AutoML and the Role of Humans in the Loop
ACM CHI'21 – Yokohama, Japan (virtual due to COVID-19)

October 2021 User ex Machina: Simulation as a design probe in human in the loop text analytics
ACM CHI'21 – Yokohama, Japan (virtual due to COVID-19)

June 2019 Automated Visualization Recommendations for Genomic Epidemiology Data
Applied Bioinformatics and Public Health Microbiology (ABPHM 2019) – Hinxton, UK
Winner: ABPHM'19 Bioinformatics Software Prize

October 2018 How to evaluate an evaluation study? Comparing and contrasting practices in vis with those of other disciplines
BELIV 18 (an IEEE VIS 2018 affiliated workshop) – Berlin, Germany

September 2018 A Method for systematically surveying data visualizations in infectious disease genomic epidemiology
ASM NGS 2018 – Tysons, USA

September 2017 Evidence-based design and evaluation of a whole genome sequencing clinical report for the reference microbiology laboratory
ASM NGS 2017 – Washington DC, USA

May 2017 Visualizing public health data for communicable disease management and control
Applied Bioinformatics and Public Health Microbiology (ABPHM 2017) – Hinxton, UK

October 2016 On regulatory and organizational constraints in visualization design and evaluation
BELIV 16 (an IEEE VIS 2016 affiliated workshop) – Baltimore, Maryland, USA

October 2016 Bridging the Gaps in Tuberculosis Research: A Public Health Data Visualization Problem
IEEE VIS Doctoral Colloquium – Baltimore, USA

Invited Talks and Panels

October 2020 The role of vis and data researchers during a public health crisis
IEEE VIS 2020

December 2018 Seeing your data: how to use data visualization for analysis and communication
American Public Health Association - webinar

September 2018 Creating explorable visualization design spaces: an example from infectious disease genomic epidemiology
Harvard Medical School, USA

June 2018 Establishing a visualization design space: A case study in infectious disease genomic epidemiology
National Microbiology Laboratory, Canada

April 2018 Visualizing Public Health Data
Dagstuhl Seminar "BioVis Crossroads" – Wadern, Germany

January 2018 An Introduction to Data Visualization
PopData British Columbia, Canada

Invited Seminars & Workshops

April 2018 **Dagstuhl Seminar – BioVis Crossroads**
Organized by: Jan Aerts (KU Leuven, Belgium); Nils Gehlenborg (Harvard University, United States of America); Elisabeta Marai (University of Illinois, United States of America); and Kay Nieselt (Universität Tübingen, Germany)
I was reinvited for the 2020 Dagshtul but declined due to the ongoing pandemic

January 2018 **AI and Public Health**
Organized by: CIHR's Institute of Population and Public Health (IPPH) and the Canadian Institute for Advanced Research (CIFAR)

Service

Editorial or Program Committees
IEEE VIS (2021 – present)
Frontiers in Bioinformatics Review Editor (2020 – present)
Bioinformatics Open Source Conference (2017 – 2019)

Organizing Committee
Papers Chair – Visualization in Data Science Symposium (2021 – Present)
IEEE VIS Organizing Committee – Vis in Practice (2021 – Present)
IEEE VIS Organizing Committee - BioVis @ VIS Chair (2018 – 2020)
BioVis Website Chair (2016 – 2018)

Reviewing
ACM CHI (2019 - present)
ACM CSCW (2022 – present)
IEEE VIS (2016 - present)
IEEE TVCG (2018 - present)
Frontiers in Bioinformatics (2020 – present)
BioKDD (2019 - 2020)
Bioinformatics Open Source Conference (2017 – 2019)
Journal of Urology (2017 – 2019)
SIGCHI DIS (2019)

Awards and Fellowships

2021 Best Paper Honorable Mention – CHI'21

2020 Best Paper Award – VDS'21

2019 ABPHM Bioinformatics Software Prize

2018 UBC Public Scholar Award ($5,000)

2018 UBC Affiliated Fellowships - Li Tze Fong Memorial Award ($25,000)

2018 CIHR Gold – National Poster Competition at CSHRF ($500)

2015 CIHR Vanier Doctoral Award ($150,000)

2015 UBC Four Year Fellowship ($72,000 + tuition)

2015 CIHR CGS-D Doctoral Award ($105,000; *I declined the award in favour of the Vanier)*

2008 CIHR-MSFHR Bioinformatic Training Program Award ($42,000)

2008 NSERC Undergraduate Summer Research Award ($8,000)

2008 CRA-W Canadian Distributed Mentors Award ($3,500)

2004 Nortel Network Entrance Scholarship ($4,000)

Summary of Qualifications

PhD in Computer Science with 7+ years of experience translating applied machine learning (ML) research to products, policy making, and innovation programs. Experience in industrial research (5 years), government (2 years), and academia.

Track record of award winning high-quality research publications accepted to top-tier conferences in computer science (ACM CHI, IEEE TVCG) and applied ML research in medicine (Nature, Bioinformatics), with over 3,500 total citations. Won competitive scholarships and fellowships including Vanier Canada Scholar, Canada's top doctoral research

Demonstrated ability to plan, design, execute, actualize, and communicate both strategic and tactical translational ML/AI research to technical and non-technical audiences

Proficiency in quantitative, qualitative, and mixed research methods. Expertise with advanced statistical data analysis, clinical study design, user research study design

Proficiency with ML/AI and Data Science tooling in Python (pandas, sklearn, nltk, hugging face, fastai, etc.), R (ggplot, tidyverse, shiny,etc.), SQL, AWS (Sagemaker)

Proficiency with Data Visualization libraries in JavaScript (Vega-lite, D3), Python (altair, streamlit) and R (ggplot, shiny), also software systems (Tableau)